Momentum: Can Past Winners Predict Future Winners?

An independent quantitative finance research project

Abstract

This project tests whether stocks with the strongest returns over the preceding twelve months, excluding the most recent month, continue to outperform in the month that follows, using a long-only 12–1 month momentum strategy applied to a fixed universe of 19 large-capitalisation US stocks over the 128 months from February 2016 to September 2026. Net of a ten-basis-point trading cost, the top-quintile winner portfolio earned an annualised return of 48.7 per cent against 25.5 per cent for an equal-weighted portfolio of the same stocks and 15.6 per cent for SPY, and its excess return over the equal-weighted universe is statistically significant (t = 2.92) and survives the removal of its single best month. The conventional test of the momentum effect, however, tells a weaker story, since the winners-minus-losers spread of 19.7 per cent a year falls short of significance (t = 1.79) and remains below conventional thresholds in every one of eight alternative specifications, while the losers themselves performed almost exactly in line with the universe. The evidence is therefore best read as showing that momentum ranking, within a universe selected with hindsight, concentrated the portfolio in the period's dominant stocks, above all NVIDIA, which was held in 79 of 128 months, rather than as robust confirmation of a cross-sectional momentum premium.

1. Introduction

The proposition that past returns contain information about future returns sits uneasily with the weak form of the efficient markets hypothesis, under which publicly observable price histories should already be reflected in prices and should therefore offer no reliable basis for earning abnormal returns. Cross-sectional momentum, the tendency of stocks that have outperformed their peers over the previous three to twelve months to continue outperforming over the following months, has nonetheless become one of the most extensively documented anomalies in empirical finance, appearing across decades, national markets and asset classes, and it now forms a standard factor in models of the cross-section of expected returns.

This project asks whether that pattern can be observed in a small universe of large, liquid US stocks over the decade to 2026, a period dominated by the concentration of market gains in a handful of technology firms. The research question is framed narrowly, as whether stocks ranked highest on their 12–1 month return go on to outperform in the subsequent month, and it is tested in two complementary ways: by comparing a long-only winner portfolio with an equal-weighted portfolio of the same stocks, which isolates the contribution of ranking from the contribution of owning the universe at all, and by measuring the return spread between winners and losers, which is the form in which the effect is conventionally tested in the academic literature.

The hypothesis is that, if momentum is present, the winner portfolio should outperform the equal-weighted universe, the losers should underperform it, and the resulting winners-minus-losers spread should be positive and statistically distinguishable from zero. The sections that follow review the relevant literature, describe the data and method, present the results and their robustness, and then argue that the pattern of findings, strong outperformance by winners alongside a weak and insignificant spread, is more informative about the interaction between momentum and hindsight-selected universes than about the existence of the momentum premium itself.

2. Literature review

The modern study of momentum begins with Jegadeesh and Titman (1993), who showed that portfolios buying US stocks in the top decile of returns over the previous three to twelve months and selling those in the bottom decile earned roughly one per cent a month over the following year, a result that could not be explained by systematic risk or by delayed reactions to common factors. Carhart (1997) subsequently added a momentum factor to the Fama–French three-factor model and found that it accounted for much of the apparent persistence in mutual fund performance, which helped to establish momentum as a standard dimension along which returns are measured, and Asness, Moskowitz and Pedersen (2013) extended the evidence to international equities, government bonds, currencies and commodities, finding momentum premia in almost every market they examined.

The convention of skipping the most recent month, which gives the 12–1 signal its name, follows from Jegadeesh (1990), who documented that returns over a single month tend to reverse in the next, a short-term effect usually attributed to bid–ask bounce and temporary price pressure, so that including the latest month in a momentum signal would partly offset the effect being measured. Novy-Marx (2012) later argued that the return

over months twelve to seven carries more predictive power than the more recent months, which suggests that the precise construction of the signal matters and that results from a single specification should be checked against alternatives.

Two broad families of explanation compete in the literature. Behavioural accounts attribute momentum to systematic errors in how investors process information, whether through conservatism and slow updating (Barberis, Shleifer and Vishny, 1998), overconfidence and biased self-attribution (Daniel, Hirshleifer and Subrahmanyam, 1998), or the gradual diffusion of news across investors who each observe only part of it (Hong and Stein, 1999); risk-based accounts, by contrast, interpret the premium as compensation for exposure to states of the world in which momentum strategies perform badly. That second interpretation draws support from Daniel and Moskowitz (2016), who showed that momentum is prone to severe crashes, typically when markets rebound sharply after a decline and the previous losers, which tend to be high-beta stocks, recover faster than the previous winners.

The practical literature adds two further cautions that bear directly on this project. Korajczyk and Sadka (2004) found that trading costs substantially reduce the profitability of momentum at scale, particularly in smaller and less liquid stocks, and Brown, Goetzmann, Ibbotson and Ross (1992) showed how restricting a sample to survivors can manufacture apparent persistence in performance where none exists, a mechanism that, as section 7 argues, is especially potent when the strategy being tested is itself a bet on persistence.

3. Data

Daily dividend- and split-adjusted closing prices were downloaded from Yahoo Finance through the open-source `yfinance` library for the period from January 2015 to September 2026 and converted to month-end prices, from which simple monthly total returns were calculated; the first year of data serves only to form the initial signal, so the evaluation period runs from February 2016 to September 2026, a total of 128 monthly observations. The risk-free rate is proxied by the yield on 13-week US Treasury bills, with the yield prevailing at the end of each month treated as the return earned on cash during the following month.

The file consists of 19 large-capitalisation US stocks drawn from technology, communication services, financials, energy and consumer sectors: AAPL, MSFT, AMZN, GOOGL, META, NVDA, AVGO, JPM, V, MA, XOM, CVX, COST, WMT, HD, NFLX, CRM, ORCL and INTC. The design originally specified 20 stocks, but price data for Adobe (ADBE)

failed to download during the run that produced these results, and the analysis was therefore conducted on the remaining 19; the code in the accompanying repository lists these 19 stocks so that the reported figures can be reproduced. SPY, an exchange-traded fund tracking the S&P 500, serves as the market benchmark.

The most important feature of this universe is the manner of its selection, since every stock in it is a large and prominent company as of 2026, chosen with knowledge of which firms became large over the sample period. This introduces a form of look-ahead selection bias that is more severe than conventional survivorship bias, because it excludes not only firms that failed but also firms that were large in 2016 and subsequently declined, and its implications for interpreting the results are taken up in sections 7 and 8.

4. Methodology

4.1 Signal

For each stock *i* and each holding month *t*, the momentum signal is the cumulative return over the eleven months from *t*−12 to *t*−2, so that the month immediately preceding portfolio formation is excluded in line with the short-term reversal evidence discussed above:

$$\mathrm{MOM}_{i,t} = \prod_{k=2}^{12} (1 + r_{i,t-k}) - 1 \tag{1}$$

The signal is formed at the close of month *t*−1 using only prices that would have been observable at that time, and the timing was verified in the code by comparing computed signals against values calculated directly from month-end prices.

4.2 Portfolio construction

At the end of each month, stocks are ranked on their signal and the top 20 per cent, which in a universe of 19 stocks amounts to three, are bought in equal weights and held for the following month, after which the portfolio is rebalanced. The bottom three stocks form a loser portfolio constructed in the same way, and the difference between the winner and loser returns gives the winners-minus-losers (WML) spread, which corresponds to a zero-cost long–short position.

In the expressions below, W and L denote the sets of winner and loser stocks selected for month t, N the number of stocks in each set, and M the number of stocks with a valid signal in that month, each carrying the subscript t where it varies over time; the portfolio returns are then:

$$N = \lfloor 0.2 \times M_t \rfloor = \lfloor 0.2 \times 19 \rfloor = 3 \tag{2}$$

$$R_t^W = \frac{1}{N} \sum_{i \in W_t} r_{i,t} \qquad R_t^L = \frac{1}{N} \sum_{i \in L_t} r_{i,t} \tag{3}$$

$$\mathrm{WML}_t = R_t^W - R_t^L \tag{4}$$

$$R_t^{EW} = \frac{1}{M_t} \sum_{i=1}^{M_t} r_{i,t} \tag{5}$$

4.3 Trading costs and turnover

Turnover is measured as the change in portfolio weights required at each rebalance, with weights allowed to drift with returns during the month so that the calculation reflects the trades that would actually be needed, and net returns deduct a cost of ten basis points per dollar traded. This cost assumption is deliberately simple but is generous for stocks of this size, whose quoted spreads are typically a small fraction of that figure; the robustness section considers how large costs would need to be to eliminate the strategy's advantage.

Formally, with w denoting the target weight of stock i in month t and c = 0.001 the cost per dollar traded, the weights carried into each rebalance after drifting with returns, the one-way turnover and the net return are:

$$\tilde{w}_{i,t-1} = \frac{w_{i,t-1}(1 + r_{i,t-1})}{\sum_j w_{j,t-1}(1 + r_{j,t-1})} \tag{6}$$

$$\mathrm{TO}_t = \frac{1}{2}\sum_i |w_{i,t} - \tilde{w}_{i,t-1}| \tag{7}$$

$$R_t^{\mathrm{net}} = R_t^W - c\sum_i |w_{i,t} - \tilde{w}_{i,t-1}| = R_t^W - 2c\,\mathrm{TO}_t \tag{8}$$

4.4 Comparisons and statistics

The winner portfolio is compared with three alternatives, each of which answers a different question. The equal-weighted portfolio of the full universe, rebalanced monthly, isolates whether ranking adds value beyond simply holding the same stocks; SPY measures performance against the broad market; and the WML spread provides the conventional test of the momentum effect. Performance is summarised by annualised geometric return, annualised volatility, the Sharpe ratio computed on returns in excess of the T-bill rate, maximum drawdown and the proportion of positive months. Statistical significance is assessed with t-tests on mean monthly returns and spreads, and with CAPM regressions of excess portfolio returns on excess SPY returns, using classical standard errors throughout.

For a portfolio with monthly return R in month t over a sample of T months, where y is the 13-week T-bill yield in per cent at the end of each month and the subscript m denotes the market return on SPY, the measures are defined as follows. The monthly risk-free rate is:

$$r_{f,t} = \frac{y_{t-1}}{100 \times 12} \tag{9}$$

Annualised geometric return and annualised volatility:

$$R_{\mathrm{ann}} = \left[\prod_{t=1}^{T}(1+R_t)\right]^{12/T} - 1 \qquad \sigma_{\mathrm{ann}} = \sqrt{12}\;\hat{\sigma}(R_t) \tag{10}$$

Sharpe ratio, computed on monthly excess returns and annualised:

$$\mathrm{SR} = \sqrt{12}\;\frac{\overline{R_t - r_{f,t}}}{\hat{\sigma}(R_t - r_{f,t})} \tag{11}$$

Maximum drawdown, the largest fall in cumulative wealth V from its previous peak:

$$V_t = \prod_{s=1}^{t} (1 + R_s) \qquad \mathrm{MDD} = \min_{t} \frac{V_t}{\max_{s \le t} V_s} - 1 \tag{12}$$

The t-statistic for a series of monthly returns or return differences d, testing whether its mean is zero:

$$t = \frac{\bar{d}}{\hat{\sigma}(d) \,/\, \sqrt{T}} \tag{13}$$

The CAPM regression, estimated by ordinary least squares:

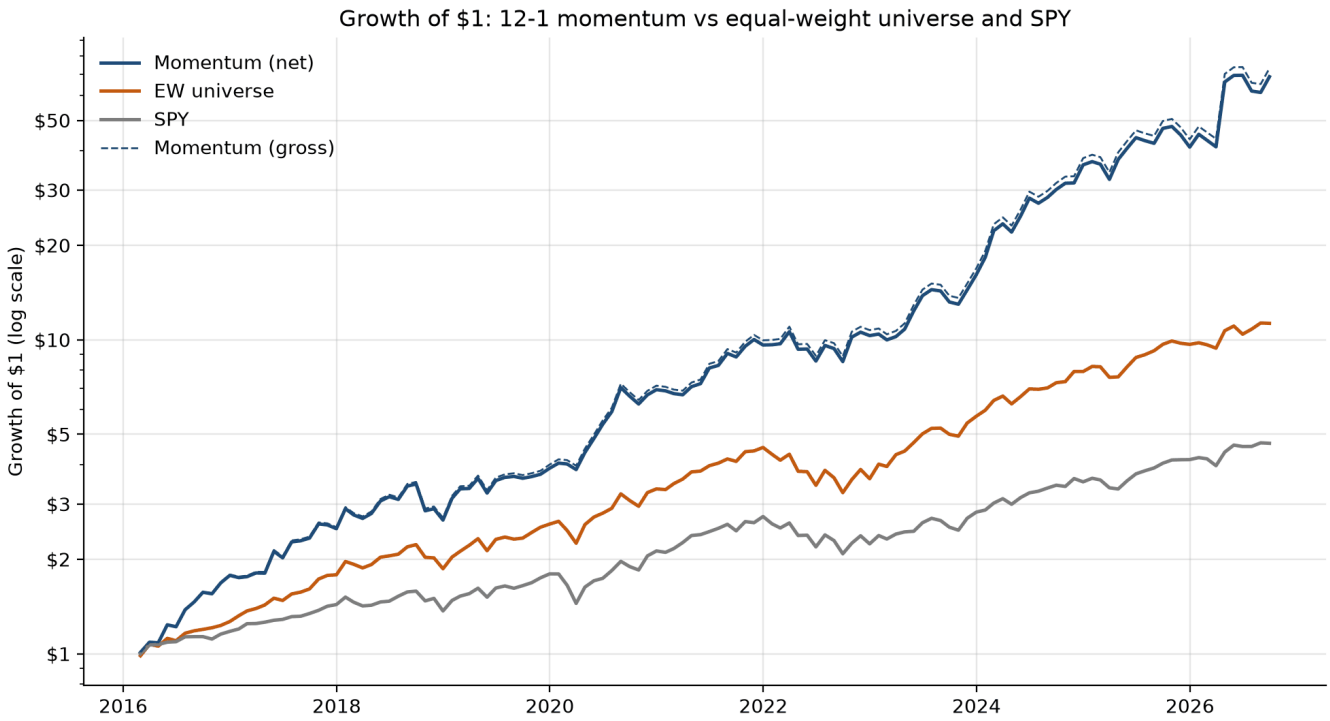
$$R_t - r_{f,t} = \alpha + \beta (R_{m,t} - r_{f,t}) + \varepsilon_t \tag{14}$$

Annualised alpha is reported as 12α.

5. Results

5.1 Headline performance

Over the 128-month sample, the momentum strategy compounded at 48.7 per cent a year net of costs, almost double the 25.5 per cent earned by the equal-weighted universe and more than three times the 15.6 per cent earned by SPY, so that one dollar invested in February 2016 grew to roughly seventy dollars by September 2026, compared with about eleven dollars in the equal-weighted universe and under five dollars in SPY (Figure 1).



This return came at a considerable cost in volatility, which at 31.3 per cent a year was nearly twice that of the equal-weighted universe, and the strategy's risk-adjusted advantage is accordingly much smaller than its raw return would suggest: its Sharpe ratio of 1.35 exceeds the 1.26 of the equal-weighted universe by a modest margin, although both comfortably exceed the 0.90 of SPY.

Measure	Momentum (net)	Momentum (gross)	EW universe	SPY	Losers	WML (gross)
Annualised return	48.7%	49.6%	25.5%	15.6%	24.6%	14.6%
Average monthly return	3.72%	3.77%	2.04%	1.31%	2.13%	1.64%
Annualised volatility	31.3%	31.3%	17.6%	15.0%	26.6%	35.8%
Sharpe ratio	1.35	1.37	1.26	0.90	0.88	0.55
Maximum drawdown	−23.2%	−23.1%	−28.2%	−23.9%	−29.4%	−48.2%
Positive months	64.8%	64.8%	69.5%	71.1%	60.2%	54.7%
Best month	60.9%	60.9%	14.6%	12.7%	25.8%	59.3%
Worst month	−17.8%	−17.8%	−11.8%	−12.5%	−23.7%	−24.6%

Table 1. Performance summary, February 2016 to September 2026 (128 months). Annualised returns are geometric; Sharpe ratios use returns in excess of the 13-week T-bill rate, except for the zero-cost WML spread.

The most revealing comparison in Table 1 is between the losers and the equal-weighted universe, since the losers earned an average monthly return of 2.13 per cent, slightly above the universe's 2.04 per cent, and an annualised return within one percentage point of it. Under the momentum hypothesis the losers should have underperformed, and the fact that they did not means that the entire WML spread is attributable to the winners, a pattern that section 7 interprets as evidence about the universe as much as about momentum.

5.2 Statistical significance

The excess return of the momentum strategy over the equal-weighted universe averaged 1.68 per cent a month, or 20.2 per cent a year, with a t-statistic of 2.92, which is significant at the one per cent level, and its excess over SPY was larger still at 29.0 per cent a year (t = 3.84). The WML spread, by contrast, averaged 1.64 per cent a month with a t-statistic of only 1.79, which falls short of significance at the five per cent level, so that the formal test of the momentum effect does not reject the null hypothesis of no spread.

Test (H_0: mean = 0)	Annualised mean	t-statistic
Momentum (net) − SPY	29.0%	3.84
Momentum (net) − EW universe	20.2%	2.92
WML (winners − losers)	19.7%	1.79

Table 2. Tests of mean monthly return differences (128 months).

5.3 Market exposure

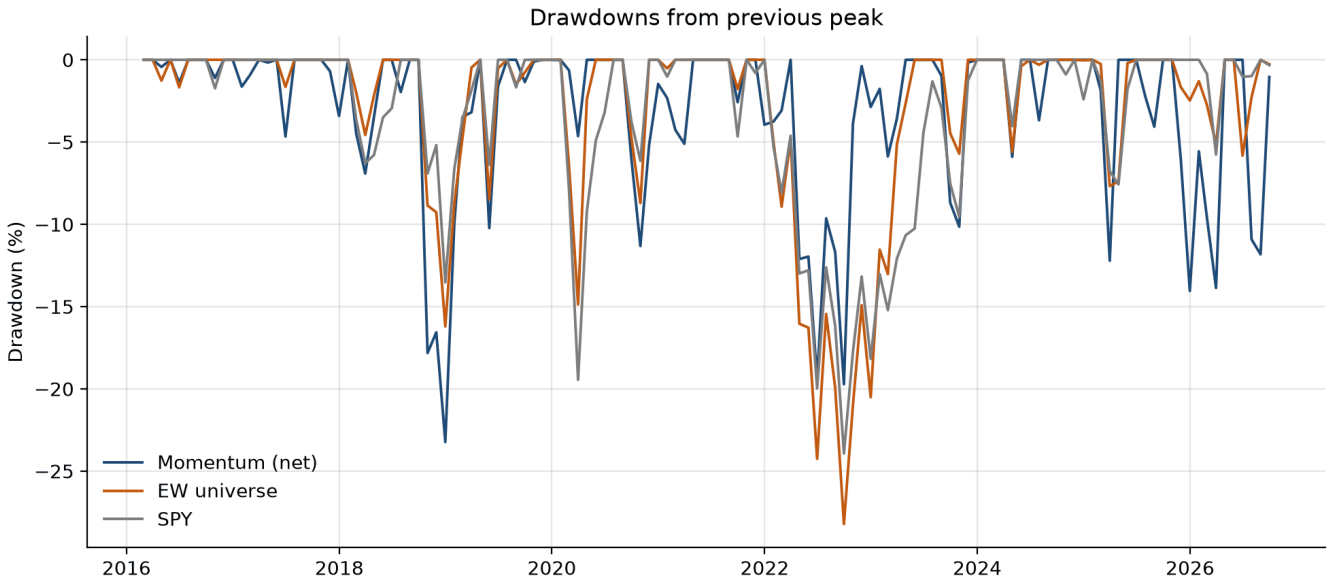
The CAPM regressions show that part of the strategy's outperformance reflects higher exposure to the market, since its beta of 1.33 implies that it moved about a third more than SPY in either direction, although a large annualised alpha of 24.6 per cent (t = 3.20) remains after this exposure is accounted for. The market explains only 40 per cent of the strategy's monthly variation, compared with 87 per cent for the equal-weighted universe, which is consistent with a portfolio of three stocks carrying a large amount of stock-specific risk. The WML spread has a beta close to zero (0.22, t = 1.03) and an alpha of 14.5 per cent a year that is not statistically significant (t = 1.28).

	Momentum (net)	EW universe	WML
Alpha (annualised)	24.6%	7.5%	14.5%
Alpha t-statistic	3.20	3.69	1.28
Beta	1.33	1.10	0.22
R-squared	0.40	0.87	0.01

Table 3. CAPM regressions of monthly excess returns on SPY excess returns.

5.4 Drawdowns and the distribution of returns

The strategy's maximum drawdown of 23.2 per cent, which occurred in late 2018, was shallower than the 28.2 per cent suffered by the equal-weighted universe during the 2022 sell-off, a result that reflects the strategy's capacity to rotate away from falling stocks (Figure 2). During 2022 the portfolio moved into ExxonMobil and Chevron as energy stocks rallied and technology stocks fell, so that its drawdown that year was smaller than the universe's, while its sharper but briefer declines in 2018 and 2025–26 reflect the concentration of a three-stock portfolio.



The distribution of monthly returns in Figure 3 shows the same concentration from another angle, since the momentum strategy's returns are more dispersed than those of either comparison portfolio and include a single extreme observation of +60.9 per cent in April 2026. That month, in which the portfolio held Alphabet, Intel and Broadcom,

coincided with a 114 per cent rise in Intel's share price following its foundry agreement with Tesla and an unexpectedly strong first-quarter result, and it illustrates how heavily a portfolio of three stocks depends on events affecting individual firms.



6. Robustness

6.1 Alternative specifications

The strategy was re-estimated with lookback periods of three, six, nine and twelve months and with winner portfolios formed from either the top 20 or the top 30 per cent of stocks, giving eight specifications in total, and the results suggest that the baseline specification is among the most favourable rather than representative of the set.

Lookback	Top fraction	Annualised return (net)	Sharpe (net)	WML annualised mean	WML t-statistic	Monthly turnover
3 months	20%	33.6%	1.17	13.9%	1.67	61.1%
3 months	30%	31.9%	1.29	11.0%	1.68	52.8%
6 months	20%	36.2%	1.16	9.0%	0.91	37.3%
6 months	30%	25.0%	0.99	0.5%	0.08	31.3%
9 months	20%	38.1%	1.20	11.9%	1.17	30.2%
9 months	30%	30.5%	1.15	5.7%	0.81	25.3%
12 months	**20%**	**48.7%**	**1.35**	**19.7%**	**1.79**	**24.5%**

Lookback	Top fraction	Annualised return (net)	Sharpe (net)	WML annualised mean	WML t-statistic	Monthly turnover
12 months	30%	37.1%	1.33	10.4%	1.38	22.2%

Table 4. Robustness to lookback period and portfolio size; the baseline specification is shown in bold. Turnover is one-way.

Every specification produced a higher raw return than the equal-weighted universe's 25.5 per cent, but only three of the eight achieved a higher Sharpe ratio than the universe's 1.26, and the WML t-statistic never reached 2, ranging from 0.08 to 1.79. The baseline 12-month, top-20-per-cent combination produced both the highest return and the strongest WML spread, which means that a researcher who had chosen any other specification would have reported markedly weaker evidence of momentum, a sensitivity that warrants caution in light of the data-mining concerns raised by Harvey, Liu and Zhu (2016).

6.2 Excluding the best month

Removing April 2026 reduces the strategy's annualised net return from 48.7 to 42.6 per cent and the WML t-statistic from 1.79 to 1.48, but leaves the t-statistic on the excess return over the equal-weighted universe almost unchanged at 2.91. The outperformance of winners over the universe is therefore not an artefact of a single month, whereas the already marginal WML result becomes weaker still.

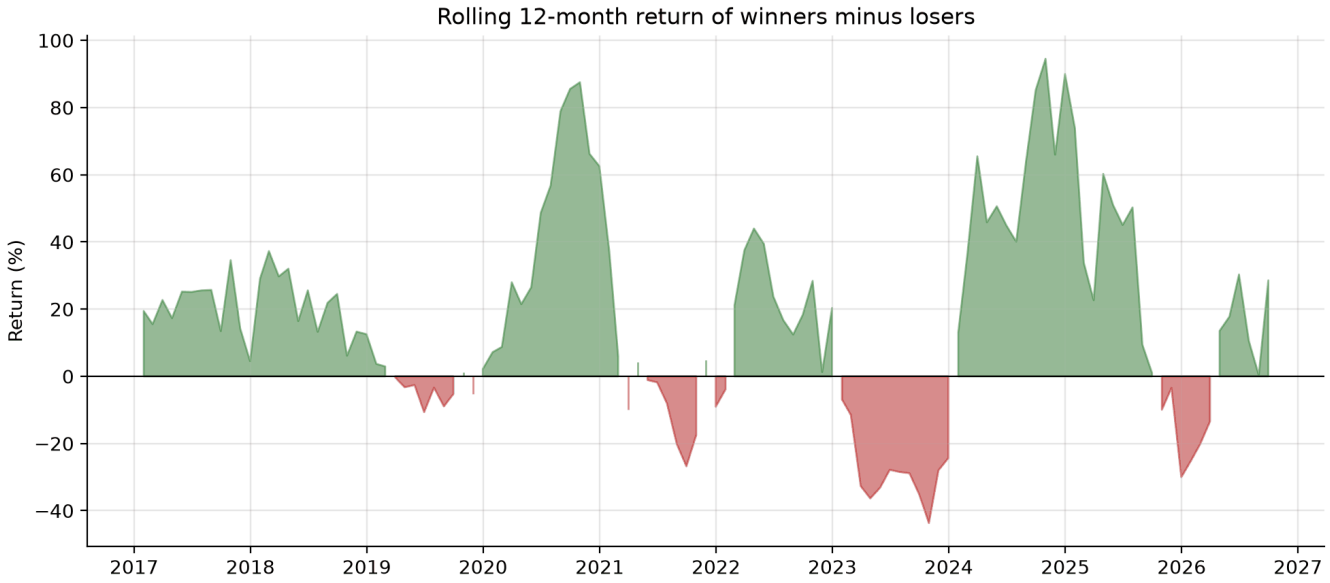
6.3 Trading costs

The strategy turned over about a quarter of its portfolio each month in the baseline specification, and the ten-basis-point cost assumption reduced its annualised return by only about one percentage point. Since roughly 49 per cent of the portfolio's value was traded each month on a two-way basis, costs would need to rise by around 340 basis points per dollar traded to eliminate the excess over the equal-weighted universe, a level far above any plausible cost of trading stocks of this size, so trading frictions do not alter the conclusions.

6.4 Time variation

The WML spread was highly unstable over time (Figure 4), with strongly positive twelve-month returns during 2017–18, 2020, early 2022 and 2024–25, interrupted by sharp reversals in 2019, late 2021, 2023 and late 2025. The deepest of these, a rolling loss of over 40 per cent in late 2023, followed the market's rebound from its 2022 decline, when the

portfolio was still positioned in the energy and defensive stocks that had led during the sell-off while the previously beaten-down technology stocks recovered, a pattern that closely matches the momentum crashes described by Daniel and Moskowitz (2016).



6.5 Concentration

The holdings record shows that NVIDIA was in the winner portfolio in 79 of the 128 months, Broadcom in 50 and Netflix in 39, so that the three most frequently held stocks accounted for 44 per cent of all portfolio positions. A large share of the strategy's return is therefore a long exposure to NVIDIA during the decade in which it became the largest company in the world, which raises the question of how much of the result reflects a general tendency for winners to keep winning and how much reflects the exceptional path of one stock.

7. Discussion

The results present an apparent tension, since the winner portfolio outperformed the equal-weighted universe by a large and statistically significant margin, yet the conventional winners-minus-losers test of momentum failed to reach significance in any specification. This tension is resolved once it is recognised that the two tests measure different things and that the first is far more exposed than the second to the way in which the universe was constructed.

The equal-weighted universe was introduced as a benchmark precisely because it shares the universe's selection bias, so that any upward distortion arising from the choice of stocks should affect the winner portfolio and the benchmark alike and cancel out of the

comparison. That reasoning holds for the level of returns but not for their distribution across stocks, because the stocks that enjoyed the largest cumulative gains over the decade, and which are present in the universe because of those gains, are also the stocks that spent the most months with high trailing returns. A momentum rule applied to a universe selected with hindsight therefore tilts systematically towards the stocks whose subsequent success determined their inclusion, and the comparison with the equal-weighted universe, while far more informative than the comparison with SPY, still overstates what the strategy would have achieved in real time. The prominence of NVIDIA, held in almost two-thirds of all months, is the clearest expression of this interaction.

The behaviour of the losers points in the same direction. In the broad markets studied by Jegadeesh and Titman (1993), a substantial part of the momentum spread comes from losers continuing to underperform, often because they are deteriorating firms on their way to distress or delisting; in a universe composed only of firms that had become dominant by 2026, no stock could have been a persistent loser of that kind, and the losers in this sample were generally temporarily weak members of an exceptionally successful group, which helps to explain why they matched the universe rather than lagging it. The absence of a significant spread is therefore partly a consequence of the same selection that inflates the winners' performance.

None of this implies that the findings are without value. The strategy's capacity to rotate into energy stocks during 2022, its shallower maximum drawdown relative to the universe and its pronounced reversal after the 2022–23 market turn are all consistent with the mechanisms described in the literature, and the 2023 episode in particular reproduces in miniature the momentum crashes documented by Daniel and Moskowitz (2016). What the sample cannot do is establish whether these patterns would have generated returns for an investor who did not know in 2016 which companies would dominate the following decade. The answer to the research question is accordingly qualified: within this universe past winners did continue to outperform, but the evidence that this reflects a momentum effect, rather than the interaction between a persistence-based rule and a persistence-selected sample, is weak.

8. Limitations

The limitations of this study follow largely from its scale, and each suggests a direction in which the analysis could be extended.

- **Selection bias.** The universe was chosen in 2026 from companies that were large and successful at that date, and, as section 7 argues, this inflates the winner portfolio's performance even relative to the equal-weighted benchmark. A point-in-time universe, such as the historical constituents of the S&P 500 at each formation date, would remove this bias.
- **Small cross-section.** With 19 stocks, each portfolio holds only three, so results are dominated by a few firms and by individual events such as Intel's rise in April 2026. Academic studies typically sort thousands of stocks into deciles.
- **Short sample.** A sample of 128 months provides limited statistical power, and the period was unusual in the degree to which market returns were concentrated in a small number of technology firms.
- **Statistical method.** The t-tests and regressions use classical standard errors, which do not allow for autocorrelation or heteroskedasticity in monthly returns; Newey–West standard errors would be more appropriate, and the strategy's returns were not tested against a full factor model.
- **Frictions and implementation.** The backtest applies a flat trading cost and ignores taxes, market impact and the cost of shorting the loser leg of the WML spread, although section 6.3 shows that realistic costs would not alter the conclusions.
- **Data quality.** Yahoo Finance data accessed through `yfinance` are adequate for educational research but are not of institutional quality, and one ticker, Adobe, failed to download, which reduced the universe from the intended 20 stocks to 19.

9. Conclusion

Applied to 19 large US stocks between 2016 and 2026, a 12–1 month momentum strategy compounded at 48.7 per cent a year net of costs and outperformed an equal-weighted portfolio of the same stocks by a statistically significant 20 per cent a year, a result that survives realistic trading costs and the exclusion of its best month. The conventional test of the momentum effect was nonetheless inconclusive, since the winners-minus-losers spread was not significant in any of eight specifications, the losers performed in line with the universe, and the baseline specification proved to be the most favourable of those tested.

The central conclusion is that, within a universe chosen with knowledge of which firms would succeed, momentum ranking concentrated the portfolio in the decade's dominant stocks, and that this interaction between a persistence-based rule and a persistence-selected sample, rather than a robust cross-sectional premium, accounts for most of the

outperformance observed. The most valuable extension of this work would be to repeat the analysis on a point-in-time universe of several hundred stocks over a longer sample, with returns evaluated against the Fama–French and Carhart factors using autocorrelation-robust standard errors, which would allow the momentum effect to be separated from the selection effects that this study has identified.

References

Asness, C. S., Moskowitz, T. J. and Pedersen, L. H. (2013). Value and momentum everywhere. *Journal of Finance*, 68(3), 929–985.

Barberis, N., Shleifer, A. and Vishny, R. (1998). A model of investor sentiment. *Journal of Financial Economics*, 49(3), 307–343.

Brown, S. J., Goetzmann, W., Ibbotson, R. G. and Ross, S. A. (1992). Survivorship bias in performance studies. *Review of Financial Studies*, 5(4), 553–580.

Carhart, M. M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52(1), 57–82.

Daniel, K., Hirshleifer, D. and Subrahmanyam, A. (1998). Investor psychology and security market under- and overreactions. *Journal of Finance*, 53(6), 1839–1885.

Daniel, K. and Moskowitz, T. J. (2016). Momentum crashes. *Journal of Financial Economics*, 122(2), 221–247.

Harvey, C. R., Liu, Y. and Zhu, H. (2016). … and the cross-section of expected returns. *Review of Financial Studies*, 29(1), 5–68.

Hong, H. and Stein, J. C. (1999). A unified theory of underreaction, momentum trading, and overreaction in asset markets. *Journal of Finance*, 54(6), 2143–2184.

Jegadeesh, N. (1990). Evidence of predictable behavior of security returns. *Journal of Finance*, 45(3), 881–898.

Jegadeesh, N. and Titman, S. (1993). Returns to buying winners and selling losers: Implications for stock market efficiency. *Journal of Finance*, 48(1), 65–91.

Korajczyk, R. A. and Sadka, R. (2004). Are momentum profits robust to trading costs? *Journal of Finance*, 59(3), 1039–1082.

Novy-Marx, R. (2012). Is momentum really momentum? *Journal of Financial Economics*, 103(3), 429–453.

Data and other sources

- Yahoo Finance historical prices, accessed through yfinance.
- On Intel's April 2026 share-price rise: How Intel stock more than doubled last month, Nasdaq / The Motley Fool, May 2026.

Appendix: Holdings frequency

Stock	Months in winner portfolio	Share of months
NVDA	79	62%
AVGO	50	39%
NFLX	39	30%
AMZN	29	23%
META	23	18%
AAPL	23	18%
XOM	20	16%
GOOGL	17	13%
MA	16	13%
MSFT	14	11%
CVX	13	10%
JPM	11	9%
WMT	11	9%
COST	10	8%
INTC	10	8%
CRM	8	6%
ORCL	8	6%
V	2	2%
HD	1	1%

Months in which each stock was among the three winners, out of 128. Each month has three positions, so the column sums to 384.

The code, data and figures that produced these results are available in the accompanying repository, in which `momentum_strategy.py` reproduces every table and `momentum_research.ipynb` presents the analysis step by step. This is an educational research project and does not constitute investment advice.